Master of Aerospace Engineering
Research Project

FPGA-Driven Simulation of the Quantum Approximate Optimisation Algorithm (QAOA)

S3 2 page Summary report

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Summary of the Research Project

Context and Motivation

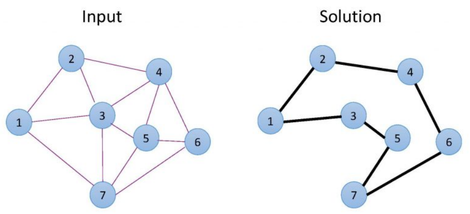

Figure 1: Travelling Salesman Problem

Variational quantum algorithms such as the Quantum Approximate Optimisation Algorithm (QAOA) are promising approaches for solving combinatorial optimisation problems. They combine a parameterised quantum circuit with a classical optimisation loop over parameters γ and β. In classical simulations, the dominant cost arises from repeatedly evaluating the quantum state evolution for different parameter values.

Field-programmable gate arrays (FPGAs) offer an attractive platform for accelerating such computations due to their parallelism and custom datapaths. This project investigates a hybrid ARM–FPGA architecture to accelerate the evaluation of a QAOA-inspired optimisation applied to a small travelling salesman problem.

Problem and Objectives

The objective is to design a hardware accelerator capable of computing the expectation value of the QAOA cost Hamiltonian for given parameters (γ, β). The project includes:

- Implementing a CPU reference QAOA simulator for a 3-city travelling salesman problem
- Designing an FPGA accelerator using High-Level Synthesis (HLS)
- Integrating the accelerator with the ARM processor through an AXI-Lite interface
- Comparing CPU and FPGA execution performance

Methodology

The travelling salesman problem (TSP) is used as the benchmark optimisation task. The distance matrix is

$$d = \begin{bmatrix} 0 & 10.0 & 4.7 \\ 10.0 & 0 & 11.0 \\ 4.7 & 11.0 & 0 \end{bmatrix}. \tag{1}$$

The problem is formulated as a QUBO and mapped to an Ising Hamiltonian. The QAOA circuit alternates between a cost Hamiltonian H_c and a mixer Hamiltonian (detailed in the long report).

$$H_C = \sum_{i,j,t} d_{ij} x_{t,i} x_{t+1,j} + P \sum_i \left(\sum_t x_{t,i} - 1 \right)^2 + P \sum_t \left(\sum_i x_{t,i} - 1 \right)^2 \tag{2}$$

The expectation value to be minimised is

$$\langle H_C \rangle = \sum_s |\psi_s|^2 H_C(s). \tag{3}$$

Two CPU implementations were developed: a grid search over (γ and β) and an optimisation using the NLopt Nelder–Mead algorithm. The FPGA accelerates the expectation evaluation while the optimisation loop runs on the ARM processor.

Hardware Architecture

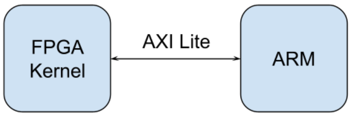

Figure 2: Overall Architecture

The accelerator was implemented using Vitis HLS targeting a Xilinx Zynq-7000 XC7Z010 device.

The quantum statevector is stored in on-chip BRAM using a custom fixed-point complex representation to reduce resource usage. The FPGA performs the quantum state evolution and expectation evaluation, while the ARM processor controls the optimisation loop. Communication between processor and accelerator uses an AXI-Lite memory-mapped interface.

Experimental Results

The CPU grid-search implementation required **678 ms** to complete the optimisation. Using NLopt reduced the runtime to 44 ms by decreasing the number of evaluations.

The ARM–FPGA implementation executes the expectation computation in programmable logic. The FPGA-accelerated grid search completed in **133 ms**, demonstrating the benefit of hardware acceleration for this computational kernel.

CPU Reference Platform	
Processor	Intel Core i7-1165G7
Cores / Threads	4 / 8
Base Frequency	2.80 GHz
Compiler	`g++` (C++17)

Table 1: CPU platform used for the software implementation.

FPGA Platform	
Board	Digilent Zybo Z7
FPGA Device	Xilinx XC7Z010
Processor	ARM Cortex-A9 (dual-core)
Frequency	650 MHz
Memory	512 MB DDR3
DSP Slices	80
Block RAM	240 KB

Table 2: Hardware platform used for the FPGA implementation.

Resource	Used	Device Capacity
LUT	11319	17600
Flip-Flops	8634	35200
DSP Slices	65	80
Block RAM	11	60

Table 3: FPGA resource utilisation for the QAOA accelerator on the XC7Z010 device.

Implementation	Optimisation Time (ms)	Evaluations
CPU Grid Search	678.49	∼2000
CPU NLopt	43.96	121
ARM–FPGA	133.15	∼2000

Table 4: Comparison of optimisation runtime across implementations.

Conclusion

This work demonstrates the integration of an FPGA accelerator in a heterogeneous ARM–FPGA system to accelerate the evaluation of a QAOA cost function. The study highlights the use of High-Level Synthesis for implementing algorithmic kernels and the practical aspects of hardware/software co-design for quantum algorithm simulation.